

Interactive Scatter Plot with Bokeh



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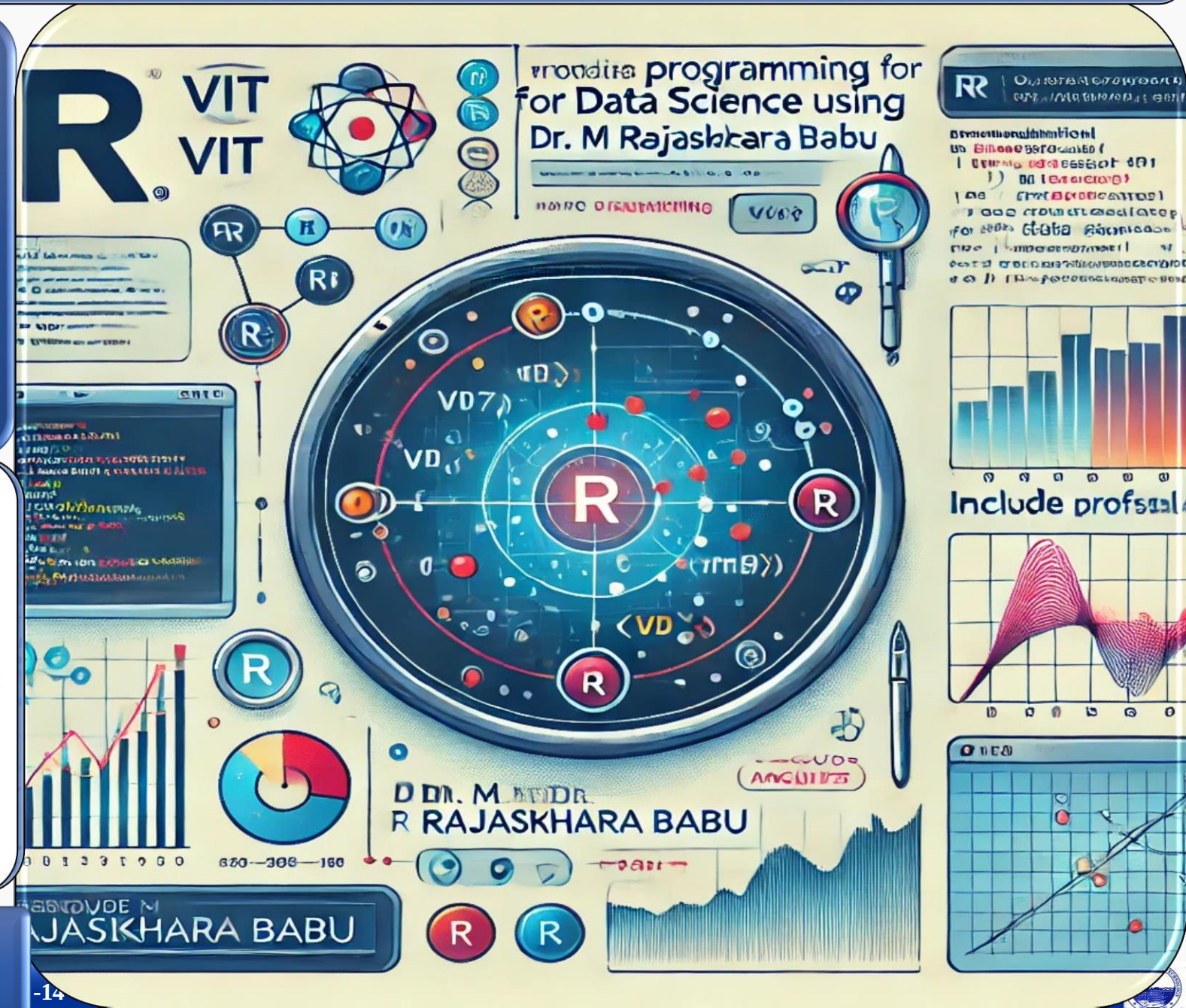


VIT[®]

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Vellore-632014, Tamil Nadu, India



What is Bokeh?

Open-Source Python Library

- Bokeh enables the creation of interactive visualizations, dashboards, and web-based graphs
- — all rendered directly in the browser.

Key Features

- **Interactive Charts**
- **Zooming & Panning**
- **Hover Tools**

Why Bokeh?

Traditional Visualization

Static graphs with no interactivity — limited exploration and fixed perspectives.

Bokeh Visualization

Interactive graphs with real-time updates, user interaction, and better data exploration.

Problem Statement

Objective

Create an interactive scatter plot showing the relationship between Petal Length and Petal Width from the Iris dataset using Bokeh.

About the Iris Dataset

One of the most popular datasets in Data Science, containing 150 samples across 3 species and 4 features.

Feature	Description
Sepal Length	Sepal length in cm
Sepal Width	Sepal width in cm
Petal Length	Petal length in cm
Petal Width	Petal width in cm

Species Classes

Setosa

Versicolor

Virginica

Install Required Libraries

```
pip install bokeh
pip install pandas
pip install scikit-learn
```

Step 1: Import Required Libraries

Begin by importing the necessary modules for plotting, data sourcing, dataset loading, and data manipulation.

```
from bokeh.plotting import figure, show
from bokeh.models import ColumnDataSource
from sklearn.datasets import load_iris
import pandas as pd
```



bokeh.plotting
Core module for
creating figures and
rendering plots.



sklearn.datasets
Provides the built-in Iris
dataset for loading.



pandas
Handles data
manipulation and
DataFrame creation.

Step 2: Load the Iris Dataset

```
iris = load_iris()
```

The `load_iris()` function from `scikit-learn` loads the complete dataset into memory, ready for processing.

150

Samples

fi

Features

3

Species

Step 3: Creating the DataFrame

Convert the raw dataset into a structured Pandas DataFrame with readable column names and species labels.

```
df = pd.DataFrame(iris.data, columns=iris.feature_names)
df.columns = ['sepal_length', 'sepal_width', 'petal_length',
              'petal_width']
df['species'] = [iris.target_names[i] for i in iris.target]
```

Easier Analysis
Named columns
make data intuitive to
work with.

Better Handling
Pandas enables
filtering, grouping,
and transformation.

Bokeh Integration
DataFrames feed
directly into Bokeh
data sources.

What is ColumnDataSource?

- A core Bokeh object that provides structured, column-oriented data for visualizations.
- It links your DataFrame directly to plot elements.

Code

```
source = ColumnDataSource(df)
```

Purpose

- Bridges data and visual elements
- Enables reactive updates
- Supports hover and selection tools

Step 5: Creating the Figure

The `figure()` function initializes the plot canvas with a title, axis labels,

```
p = figure(  
    title="Petal Length vs Petal Width",  
    x_axis_label="Petal Length (cm)",  
    y_axis_label="Petal Width (cm)",  
    width=700,  
    height=500,  
  
    tools="pan,wheel_zoom,box_zoom,reset,save"  
)
```

Title

X & Y Axes

Canvas Size

Tools

Step 6: Adding the Scatter Plot

```
p.scatter(  
    x='petal_length',  
    y='petal_width',  
    source=source,  
    size=8,  
    color="green",  
    alpha=0.7  
)
```

What Each Parameter Does

Parameter	Purpose
x, y	Column names for axes
source	Links ColumnDataSource
size	Point diameter in pixels
alpha	Transparency (0–1)

Each point on the plot represents one flower observation.

Step 7: Interactive Tools



Pan

Click and drag to navigate the plot.



Wheel Zoom

Scroll to zoom in and out on data points.



Reset

Return to the original view instantly.



Save

Export the plot as a PNG image.

Step 8: Display the Plot

The Final Command

```
show(p)
```

Renders the complete visualization in your default web browser as an HTML file.

What You'll See

- Petal Length on the X-axis
- Petal Width on the Y-axis
- 150 green scatter points
- Full interactive toolbar

 **No server required — Bokeh outputs a standalone HTML file.**

Understanding Scatter Plots

A scatter plot displays the relationship between two numerical variables. Each point corresponds to a single observation — in this case, one iris flower.

X-Axis → Petal Length (cm)

The independent variable — the feature we use to explain variation.

Y-Axis → Petal Width (cm)

The dependent variable — the feature we observe as a response.



Interpreting the Plot



Positive Correlation

Petal width increases as petal length increases — a clear positive trend across all 150 observations.



Visible Clusters

Different clusters in the plot likely correspond to different species — Setosa, Versicolor, and Virginica.



Key Insight

Flowers with longer petals generally have wider petals, suggesting petal dimensions are strongly correlated features.

Adding the HoverTool

```
from bokeh.models import HoverTool

hover = HoverTool(
    tooltips=[
        ("Petal Length",
"@petal_length"),
        ("Petal Width",
"@petal_width")
    ]
)

p.add_tools(hover)
```

Benefit

When a user hovers over any data point, a tooltip displays the exact Petal Length and Petal Width values for that flower — making exploration precise and intuitive.

📱 **HoverTool is one of Bokeh's most powerful features for communicating data details on demand.**



Biology

Species classification based on morphological measurements.



Healthcare

Visualizing medical measurements and patient data relationships.



Agriculture

Analyzing plant growth patterns and crop feature correlations.



Machine Learning

Exploratory data analysis and feature relationship discovery.

Apply what you've learned by creating a new interactive scatter plot using different Iris features.

New Feature Pair

Axis	Feature
X-Axis	Sepal Length
Y-Axis	Sepal Width

Tasks

1. Load the Iris Dataset
2. Create a DataFrame
3. Generate the Scatter Plot
4. Add a Hover Tool
5. Interpret the relationship

Full Python Code

```
from bokeh.plotting import figure, show
from bokeh.models import ColumnDataSource,
HoverTool
from sklearn.datasets import load_iris
import pandas as pd

iris = load_iris()
df = pd.DataFrame(iris.data,
columns=iris.feature_names)
df.columns =
['sepal_length', 'sepal_width', 'petal_length', 'p
etal_width']
df['species'] = [iris.target_names[i] for i in
iris.target]

source = ColumnDataSource(df)
p = figure(title="Petal Length vs Petal Width",
x_axis_label="Petal Length (cm)",
y_axis_label="Petal Width (cm)",
width=700, height=500,
tools="pan,wheel_zoom,box_zoom,reset,save")
p.scatter(x='petal_length', y='petal_width',
source=source, size=8, color="green",
alpha=0.7)
hover = HoverTool(tooltips=[("Petal
Length", "@petal_length"),
("Petal Width", "@petal_width")])
p.add_tools(hover)
show(p)
```

01

Bokeh Library

Open-source interactive visualization for Python.

02

Iris Dataset

150 samples, 4 features, 3 species — a classic ML dataset.

03

DataFrame & ColumnDataSource

Structured data pipeline from pandas to Bokeh.

04

Scatter Plot & Tools

Pan, zoom, reset, save, and hover for full interactivity.

05

Relationship Analysis

Positive correlation between petal length and petal width confirmed.